

Vidhisha Katariya

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EDUCATION

K.J. SOMAIYA INSTITUTE OF TECHNOLOGY , Mumbai

Bachelor of Technology in Electronics and Telecommunication, Honors in Data Science

CGPA – 9.1

2021 – 2025

EXPERIENCE

LTM

July 2025 – Present

Software Engineer

Mumbai, IN

- Engineered Python-based AI automation pipelines powered health-check workflows, cutting manual monitoring effort by 25% and improving operational efficiency across production systems.
- Designed real-time observability solutions using SQL and Grafana for anomaly detection applied data-driven insights to reduce incident response time by 20% for critical systems.
- Applied prompt engineering best practices to iteratively refine AI outputs, monitoring accuracy, relevance, and completeness metrics to ensure high-quality AI deployments.
- Collaborated cross-functionally with product and engineering teams to define AI solution requirements, produce standardized documentation, and ensure seamless integration into existing workflows.
- Delivered data driven reports translating complex AI outputs into actionable business recommendations, reducing potential downtime risk by 40%.

Active Spatial Sciences Research Group

July 2024 - July 2025

Research Intern

Remote, IN

- Built an end-to-end ML analytics platform in Python (Flask, NumPy, Folium, GEE APIs) serving 100+ data points with real-time trend analysis, pattern identification, and interactive filtering capabilities.
- Constructed and evaluated ML models on 50+ GB satellite datasets over 6+ years; designed preprocessing pipelines and time-series models in Python, improving data quality metrics by 30%

Godavari Biorefineries

May 2023 – July 2023

Data Analyst Intern

Karnataka, IN

- Applied Python-based outlier detection ,data preprocessing and quality checks to clean and validate geospatial datasets (500+ acres), improving mapping accuracy by 20%.
- Analyzed agricultural and spatial datasets using Excel and Earth Engine; translated analytical insights into actionable business recommendations for management.

PROJECTS

Interactive Dashboard & Analytics Platform for Geospatial Intelligence

Python, NumPy, Plotly, Machine Learning, Flask, Pandas, Folium

- Engineered a geospatial ML pipeline processing 1M+ USGS satellite data points with 85% accuracy.
- Built an interactive dashboard with advanced filtering and trend analysis features.
- Published quantitative research in [Springer journal](#).

Semantic Book Recommender

Python, LangChain, HuggingFace, Chroma Vector Store, Jupyter Notebooks, Gradio, Pandas

- Built an end-to-end semantic search and recommendation system over a 5,000+ book dataset using LangChain embeddings and Chroma vector store; applied text chunking and tone/mood categorisation enabling natural-language queries surface relevant titles beyond keyword matching.
- Designed a multi-stage NLP pipeline across data cleaning, sentiment analysis, zero-shot text classification and vector search.
- Deployed an interactive [Gradio](#) dashboard quering books by mood, category or description. emotion-tagged metadata to enhance recommendation personalisation and contextual accuracy.

TECHNICAL SKILLS

Languages: Python, C/C++, SQL, JavaScript, Java, HTML/CSS, TypeScript

AI / GenAI: Prompt Engineering, LLM Optimization, NLP, RAG, LangChain, Vector Databases, AI Model Evaluation

Frameworks & Libray: PyTorch, FastAPI, Flask, Pandas, NumPy, Matplotlib, Tensorflow, ETL, React.js

Cloud & Tools: AWS, Google Cloud Platform (GEE), Tableau, Power BI, GitHub, Figma, Grafana, Excel

Soft Skills: Cross-functional collaboration, technical communication, iterative problem-solving, research & documentation

EXTRACURRICULAR

- AWS AI/ML Scholar 2024** — Selected among top 2,500 candidates globally.
- Technical Content Writer, GeeksforGeeks** — Authored 30+ technical articles on algorithms and data structures to explain complex quantitative concepts, reaching 50K+ readers.
- Mentor at GirlScript Summer of Code 2024** — Led opensource teams.
- State Board Top 1% (Class XII, 95.6%).
- Senior Volunteer** at National Service Scheme KJSIT— Led 70+ community service initiatives demonstrating leadership and collaboration
- Open Source Contributor, Social Winter of Code 2023** — Contributed to algorithm optimization projects.